MeanFlow, from scratch

Learn the average, not the instant

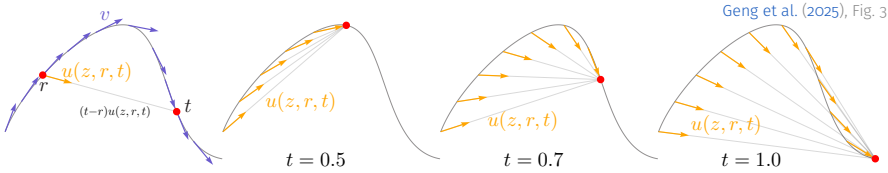
scan, then Run all in Colab

Juntang Wang · Hacker

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Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, Kaiming He.
Mean Flows for One-step Generative Modeling. arXiv 2505.13447

Method, as implemented



Geng et al. (2025), Fig. 3

paper, Alg. 1

```
t, r = sample_t_r()
e = randn_like(x)
z = (1 - t) * x + t * e
v = e - x
u, dudt = jvp(fn, (z, r, t), (v, 0, 1))

u_tgt = v - (t - r) * dudt
error = u - stopgrad(u_tgt)
loss = metric(error)
```

ours, `meanflow.py` (abridged)[1]

```
t, r, w = self.sample_t_r(batch_size, device)
e = torch.randn_like(x)
z = (1 - t_) * x + t_ * e
v = e - x
_, dudt = jvp(model, (z, t, r), (v_c, 1, 0))
u_p, v_p = model(z, t, r)
u_tgt = v_hat - (t_ - r_) * dudt
mf_loss = adaptive_l2_loss(u_p - stopgrad(u_tgt))
return mf_loss + fm_loss
```

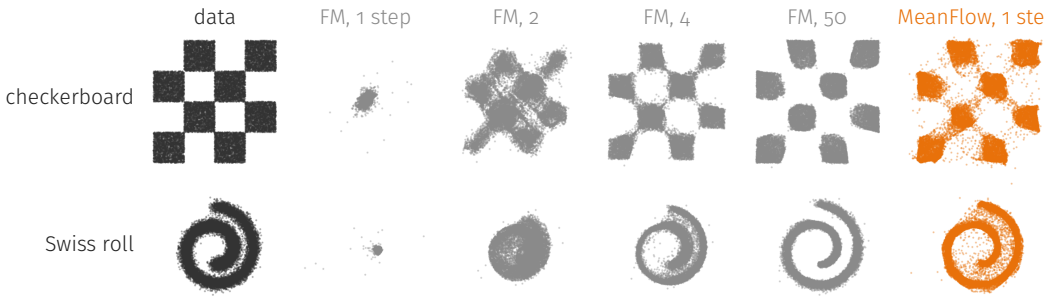
`v_c`: the model's own v head, not $e - x$; `fm_loss` trains it
`v_hat` $= v + (1 - 1/w)(v_c - v_{uc})$: v with CFG folded in
[1]adapted from `github.com/haidog-yaqub/MeanFlow`

Data

dataset	n	d	what
Synthetic			
checkerboard (Grathwohl et al., 2019)	∞	2	point on 8 squares
Swiss roll (Sohl-Dickstein et al., 2015)	∞	2	point on a thin spiral
Real-world			
MNIST (LeCun et al., 1998)	60,000	$1\times28\times28$	digit, 10 classes
Fashion-MNIST (Xiao et al., 2017)	60,000	$1\times28\times28$	clothing, 10 classes
Real-world, big[1]			
ImageNet (Deng et al., 2009)	1,281,167	$3\times256\times256$	photo, 1000 classes

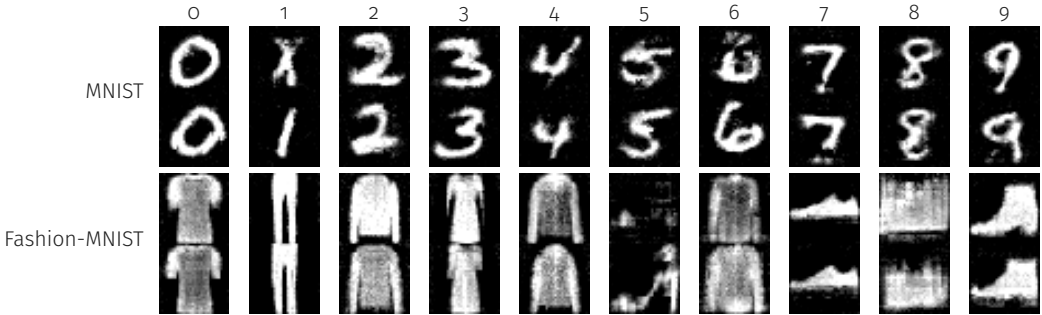
[1] training configuration: MF-XL/2, 240 epochs, batch 256, $4\times32\times32$ VAE latents, $8\times$H100. The other datasets: a single RTX 5080 / Colab

Results



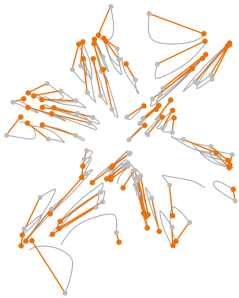
same starting noise in every panel; FM = the v head, integrated

Results (contd.)



MeanFlow, 1 step, two samples per class
Fashion: T-shirt, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, ankle boot

The jump is the average of the path



v head, 100 steps vs *u* head, 1 jump

jump vs path end: **0.08** (shuffled 4.34)



top *v* head, 100 steps · bottom *u* head, 1 step

per-pixel gap: **0.08** (shuffled 0.21)

$$u = \frac{1}{t-r} \int_r^t v \, d\tau$$

one network, two heads, same starting noise

Break the JVP, break the jump



$(v, 0, 1)$, correct

paper FID 61 · ours 0.65



$(v, 1, 1)$

paper FID 138 · ours 0.60



$(v, 0, 0)$

paper FID 268 · ours 0.24



$(v, 1, 0)$

paper FID 329 · ours 0.29

JVP tangent for $(\partial_z u, \partial_r u, \partial_t u)$

paper: Tab. 1b, ImageNet FID ↓ · ours: checkerboard overlap ↑ · same order except the two worst



Thank you

one MeanFlow step, ImageNet class 281 (tabby), 1 of 8 seeds; model by zhuyu-cs/MeanFlow

→ back to Colab: the live run

Setup

	Synthetic	**Real-world**	**Real-world, big**
code	ours	ours	zhuyu-cs
network	MLP 5×512, Fourier z	DiT, 4 layers, width 192	DiT-XL/2, 28 layers
input	$1\times1\times2$ (2D point, reshaped)	$1\times32\times32$	$4\times32\times32$ latent
params	1.1M	3.1M, patch 4×4	677M, patch 2×2
loss	`meanflow` mode, $p = 0.75$	`meanflow` mode, $p = 1$	MeanFlow + CFG
t, r sampler	logit-normal$(-2, 2)$	logit-normal$(-0.4, 1)$	—
JVP	`torch.func.jvp`, tangent from the v head		—
CFG	none	$w \in [1, 7.5]$, sampled at 4.25	folded in
optimizer	AdamW, lr 10^{-3}, cosine; batch 1024 / 128		batch 256
training	20,000 steps	3,000 steps	240 epochs
time	94 s each (RTX 5080)	131 s each (RTX 5080)	$8\times$H100

References

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.

Will Grathwohl, Ricky T. Q. Chen, Jesse Bettencourt, Ilya Sutskever, and David Duvenaud. FFJORD: Free-form continuous dynamics for scalable reversible generative models. In *International Conference on Learning Representations (ICLR)*, 2019.

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International Conference on Machine Learning (ICML)*, 2015.

Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.